Hybridizing Metaheuristics with Generative AI Techniques for Selected Graph Problems

Research project for Bosnia and Herzegovina–Austria bilateral cooperation

1. Background and motivation

Combinatorial optimization is a cornerstone of modern decision support in logistics, energy systems, telecommunication networks, transportation, and data-driven planning. Many such problems are defined or can be modeled on graphs. Examples of classical graph problems are finding a maximum independent set (MIS), a maximum cut (MaxCut), a densest subgraph of certain size, dominating sets, and diverse vehicle routing problems. These problems are typically NP-hard and can be formulated as large-scale combinatorial search tasks in which the quality of the solution depends on complex structural dependencies and constraints. Additionally, uncertainty or not yet known information in dynamic scenarios often plays important roles in real applications. Exact methods like mathematical programming techniques are powerful and provide quality guarantees but often become intractable on large or dynamic instances. (Meta-)heuristic approaches typically scale better and are in practice often favored, although they usually do not provide quality guarantees and usually rely heavily on hand-crafted operators, parameter tuning, and domain-specific engineering (Blum and Roli, 2003; Glover and Kochenberger, 2003).

Large Neighborhood Search (LNS) and **Construct, Merge, Solve and Adapt** (CMSA) (Blum et al., 2016) are two well-known and widely used metaheuristics, which are known for their flexibility. Both build on the fundamental idea to iteratively derive from the original problem smaller subproblems that are easier to handle by either an exact or heuristic technique. Solutions to these subproblems are then expected to lead to better and better solutions for the original problem. LNS realizes this principle by **destroy-and-repair** operators, which define subproblems by removing solution components or unassigning variables and solve the subproblem of augmenting the solution to a complete feasible one again, respectively. In contrast, CMSA iteratively creates diverse candidate solutions, typically by fast randomized heuristics, collects their components in a **component pool**, and solves a merging subproblem that asks for a best solution just consisting of pool components; after obtaining a solution, the component pool is updated and the process iterated. LNS and CMSA can be seen to be orthogonal approaches in the way they derive subproblems.

Besides classical optimization techniques, machine learning and artificial intelligence approaches are nowadays gaining substantial importance in combinatorial optimization. Data-driven models can identify structural patterns, approximate promising regions of the search space, and support decisions such as which components are promising to remove from or add to candidate solutions, which variables to branch on, how to be well prepared for unknown future aspects in dynamic/stochastic problems, and how to control the search process adaptively. Recent work in learning-enhanced combinatorial optimization is manifold; for an overview see, e.g., (Bengio et al., 2021).

The proposed project builds on these developments by investigating the **integration and combination of advanced generative AI models with LNS and CMSA on selected classes of graph problems**. In particular, **Denoising Diffusion Models** (DDMs), **Generative Flow Networks** (GFlowNets), (Bengio et al., 2023), and **Large Language Models** (LLMs) will be used for (a) creating large sets of meaningful diversified candidate solutions in CMSA, (b) choosing sets of solution components to destroy in LNS, and (c) augmenting partial solutions in the repair step of LNS. Moreover, the project also investigates the **integration of classical Column Generation** (CG) in CMSA and the **hybridization of LNS and CMSA**, in which their principles are combined. Last but not least, learning-based approaches will also be used to adaptively guide the search process in these metaheuristics, for example by selecting most promising operators or adapting strategy

parameters. The project hypothesis is that implicit knowledge encoded in graph-aware neural models and generative architectures can guide the heuristic search at least as or more effectively than purely hand-crafted problem-specific operators. Our recent research on diffusion-based evolutionary search, GFlowNet-based destroy-set generation, and learning-enhanced CMSA provides a strong scientific foundation for this bilateral project (Akbay et al., 2025; Bresich et al., 2026a; Reixach and Blum, 2024; Reixach and Blum, 2025; Salvà Soler and Raidl, 2026; Wustinger et al., 2026).

Developed techniques will on the one hand be benchmarked on classical node-selection problems on graphs such as MIS, MaxCut and the densest k-subgraph problem, but on the other hand also be specifically extended for two structurally more challenging applications: the **Minimum Consistent Subset Problem (MCS)** and the **Dynamic Electric Autonomous Dial-a-Ride Problem** (DynEADARP). The project is designed as a genuinely bilateral initiative in which the two partner teams contribute complementary methodological and application expertise. The Austrian partner will primarily provide expertise in LNS, denoising diffusion on graphs, and GFlowNet-based search as well as in vehicle routing, while the Bosnia and Herzegovina partner will primarily contribute expertise in CMSA, graph-based combinatorial optimization, mathematical formulations, benchmark design, and learning-assisted component generation.

2. State of the art and research gap

The field of learning-enhanced optimization has developed rapidly in recent years. Methodological surveys show that machine learning can support branching, solution candidate construction, algorithm configuration, and other decisions within exact and heuristic solvers. Within metaheuristics, data-driven control has been investigated for neighborhood selection, operator scheduling, solution construction, and search-space reduction. In this context, CMSA and LNS are particularly attractive because their performance depends strongly on how promising partial solutions, components, and destroy sets are created/selected (Reixach and Blum, 2024; Reixach and Blum, 2025; Bresich et al., 2026a).

Recent advances in generative artificial intelligence provide more expressive alternatives to classical machine learning techniques. GFlowNets learn stochastic policies for constructing structured objects with probabilities related to their reward. Unlike classical reinforcement learning, which aims to learn a policy for constructing a single structure that maximizes reward, GFlowNets are particularly suited for sampling *diverse* high-reward structures (Bengio et al., 2023). This property is highly relevant for the integration in CMSA and LNS: Both require not just a single strong prediction but a multitude of structurally different promising candidates. GFlowNets are therefore natural tools for generating diverse candidate subsets, partial solutions, and destroy sets in graph-based optimization (Bresich et al., 2026a).

Denoising diffusion models, well known for their impressive results in image and video generation, also have been adapted to combinatorial optimization (Sun and Yang, 2023). Discrete and graph-based diffusion approaches progressively transform noisy structures into high-quality candidate solutions and can be combined with classical search to improve exploration and diversification. Recent work has demonstrated the potential of diffusion-based evolutionary and adaptive search for hard graph problems, including the MIS and a generalization of the domination problem (Salvà Soler and Raidl, 2026; Wustinger et al., 2026). Similarly to GFlowNets, diffusion approaches are able to create large sets of diversified promising candidate structures.

LLMs have recently emerged as a conceptually different tool for combinatorial optimization. Using LLMs directly as optimizers (Yang et al., 2023) works only for small instances. More promising is their use as generators of heuristic code within an evolutionary loop: FunSearch (Romera-Paredes et al., 2024) found new cap set constructions and improved online bin packing heuristics, Evolution of Heuristics (Liu et al., 2024) and ReEvo (Ye et al., 2024) automatically designed heuristics and metaheuristic components for routing, scheduling, and packing problems, and AlphaEvolve (Novikov et al., 2025) improved best-known results for several mathematical and algorithmic problems. Here, the LLM generates candidate heuristics offline, which

are then evaluated and selected automatically. Most of this work, however, targets comparatively simple heuristics; generating specialized components for hybrid metaheuristics such as LNS and CMSA has hardly been investigated.
Despite many recent advances, important more general methodological questions remain open. Existing approaches typically focus on individual interactions between machine learning and optimization, such as branching, construction, neighborhood selection, destroy-set generation, or search-space reduction. The systematic integration of several learned decisions within modular hybrid metaheuristics remains relatively underexplored.

3. Scientific objectives

The project pursues the following objectives:

(i) To develop graph-aware DDM and GFlowNet-based learning mechanisms for diversified candidate solution creation in CMSA, destroy-set generation and solution augmentation in LNS, targeted towards the selected classes of graph problems.
(ii) To investigate the hybridization of LNS and CMSA as well as CG in a joint framework, benefiting from synergy.
(iii) To investigate the use of LLMs as generators of lower-level heuristics, scoring functions, and search-control rules that can be tested and integrated within a reproducible algorithmic framework.
(iv) To validate these methods on benchmark graph problems including the classical MIS, MaxCut, and densest k-subgraph problems as well as the structurally more demanding MCS and the DynEADARP.

The overall goal is to create a hybrid algorithmic pipeline that fuses the robustness of exact subroutines and metaheuristics with the expressiveness and generalization power of modern generative, neural techniques.

4. Proposed methodology and work packages

The project will be organized around a set of interconnected work packages that mirror the scientific agenda of the proposal.

WP1. Modular hybrid optimization framework

This work package will establish a modular optimization framework centered on CMSA- and LNS-based search. Constructive processes, component generation, destroy/repair mechanisms, and reduced-subproblem optimization will be exposed through common interfaces so that learned policies and generative models can be evaluated under comparable conditions. Exact as well as heuristic subproblem solvers will be used as appropriate, depending on the concrete problem and instance characteristics. The aim is not to force all techniques into a single monolithic algorithm, but to create a common experimental backbone for studying how learning can improve the generation and control of reduced search spaces (Akbay et al., 2025; Reixach and Blum, 2024; Reixach and Blum, 2025).

WP2. DDM and GFlowNet models for destroy-set and candidate generation/augmentation

This work package develops DDM and GFlowNet approaches to generate structured destroy sets, candidate subsets, and component pools that are more informative than randomly sampled neighborhoods. For LNS, the learned model will propose destroy sets that target critical vertices, edges, or components capable of triggering strong repair phases. For CMSA, the model will generate candidate components or partial solutions whose inclusion is likely to improve the objective value. For the repair step of LNS, the generative models will be conditioned on the partial solution and will propose completions that serve either directly as repaired solutions or as warm starts and variable fixings for exact repair. Training will be done primarily offline on

representative problem instances, using reward signals derived from the objective function for GFlowNets and dedicated techniques for obtaining meaningful training data from high-quality solutions for DDMs. Generalization from smaller training instances to larger and structurally different test instances will be studied explicitly. A key research challenge is to combine diversity, feasibility, and reward-aware generation in a way that is compatible with exact repair and search intensification. This direction is aligned with recent work on diffusion-based search and GFlowNet-guided neighborhood generation (Bengio et al., 2023; Bresich et al., 2026a; Salvà Soler and Raidl, 2026; Wustinger et al., 2026).

WP3. LLMs for lower-level heuristic synthesis

This work package explores LLMs as offline algorithm-design assistants. LLMs will be used to propose candidate lower-level heuristics, repair operators, scoring functions, neighborhood definitions, and search-control rules from a high-level specification. Generated routines will be translated into executable code, automatically filtered by feasibility and safety checks, and evaluated using objective optimization performance. The goal is not to replace the solver or to invoke an LLM at every search iteration, but to create a reproducible human-AI co-design framework in which LLMs propose candidate algorithmic ingredients and the optimization framework selects them according to empirical evidence. This exploratory direction is motivated by recent work on LLM-based algorithm discovery and heuristic evolution (Romera-Paredes et al., 2024; Liu et al., 2024; Yang et al., 2023).

WP4. Application 1: Minimum Consistent Subset Problem (MCS)

The MCS is defined as follows. Let $G=(V,E)$ be a connected graph in which each vertex has one of k colors. A subset $S \subseteq V$ is consistent if, for every vertex $v \in V$, the vertices of S that are closest to v contain at least one vertex having the same color as v. The MCS asks for a consistent subset of minimum cardinality (Banerjee et al., 2018; Dey et al., 2023).

This problem provides a natural model of **representative or prototype selection on colored graphs**. Vertices may represent objects, observations, or agents; graph distances describe relationships or similarity; and colors represent classes or categories. Application contexts include graph-based classification, representative selection, network summarization, and networked data analysis. MCS is especially suitable for learning-enhanced optimization because the relevance of a vertex depends simultaneously on its color, its structural position in the network, shortest-path relations, and the current set of selected representatives.

The project will benchmark exact formulations, CMSA- and LNS-based hybrid methods, and learned component-generation strategies on synthetic and real-world instances and compare them with relevant state-of-the-art approaches from literature.

WP5. Application 2: Dynamic Electric Autonomous Dial-a-Ride Problem (DynEADARP)

The DynEADARP, as defined in Tomandl et al. (2026), concerns a fleet of electric autonomous vehicles by which incoming travel requests of users consisting of pickup and delivery locations as well as time windows shall be fulfilled. In contrast to classical taxi services, users may share rides as long as detours do not get too long. This problem extends the classical dial-a-ride problem in particular by the need to plan suitable charging stops. Moreover, we consider the stochastic dynamic (online) variant in which requests only become available over time and the planning has to be done in a way to be well prepared for the expected future. This problem provides a realistic setting in which learning-based anticipatory decisions may be particularly valuable (Baty et al., 2024). The project will first focus on the simpler static variant, for which the Austrian partner has already developed a leading LNS (Bresich et al., 2026b; Bresich et al., 2026c). Moreover, the group also already proposed a simple reinforcement-learning-based extension of this LNS towards the DynEADARP, which will be used as a baseline (Tomandl et al., 2026). Building on this, the learned destroy-set and repair models will be extended to the dynamic variant, where they are additionally conditioned on

information about expected future requests. The two principal application domains deliberately represent complementary settings. Their joint study will therefore test whether the proposed learning mechanisms transfer across substantially different optimization problem types.

5. Experimental evaluation and risk contingency

The proposed learning-enhanced methods will be evaluated against conventional CMSA and LNS baselines, manually designed operators, exact approaches where computationally feasible, and relevant recent methods from the literature. Performance will be assessed using objective value or optimality gap, runtime, time-to-target, anytime performance, robustness over repeated runs, diversity of generated candidates, reduced-subproblem size, and generalization to larger or structurally different instances. Training, inference, and optimization times will be reported separately in order to assess whether the computational cost of learning is justified and amortized across repeated problem solving.

The modular design limits the scientific risk of individual learning components. If GFlowNet- or diffusion-based generation does not perform well, the same framework will support simpler neural or graph-based scoring models. If online neural adaptation proves unstable or too expensive, lightweight contextual-bandit or adaptive-probability mechanisms will be used. LLM-based heuristic synthesis is explicitly exploratory and is not a critical dependency for successful project completion.

6. Expected results and impact

The project will generate several tangible outputs. First, it will produce a modular hybrid optimization framework that combines classical search with graph-aware learning, generative models, and decision-focused control. Second, it will contribute with better solution approaches for the considered application scenarios, i.e., MCS and DynEADARP. Third, it will create reusable algorithmic modules for destroy-set generation, candidate construction, and adaptive heuristic control. Fourth, it will produce open and reproducible software components aligned with the planned work packages.

The expected scientific impact lies in demonstrating how generative AI and machine learning can be embedded within robust optimization algorithms rather than used only as isolated end-to-end models. The project will generate methodological knowledge on when learned search components provide measurable benefits, how well they generalize, and how their computational cost should be balanced against optimization gains. We plan to publish at least two joint papers in high-ranking journals, such as Computers & Operations Research, and two more joint conference paper contributions, for example at CPAIOR.

7. Bilateral cooperation and division of responsibilities

The project deliberately couples two complementary algorithmic traditions and two distinct application settings. The Austrian partner will lead work on LNS, generative optimization, diffusion and GFlowNet-based search, and DynEADARP. The Bosnia and Herzegovina partner will lead work on CMSA, graph combinatorial optimization, mathematical formulations, MCS, and learning-assisted component generation.

The partners will jointly design the common interfaces between learning and optimization, compare different learning strategies, study cross-problem and out-of-distribution generalization, and develop the LLM-assisted heuristic-synthesis pipeline. Two mutual research visits will be organized per year around concrete joint tasks—algorithm integration, model training, benchmark generation, computational experimentation, and paper preparation—so that mobility directly supports scientific deliverables.

The collaboration will also support co-supervision of graduate students and early-stage researchers, shared software development, joint publications, and preparation of follow-up European proposals at the intersection of operations research and artificial intelligence.

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